

**HAI LIU, Ph.D.**

Assistant Professor  
School of Electrical, Computer and Energy Engineering  
Ira A. Fulton Schools of Engineering | Arizona State University  
Tempe, AZ 85287  
Email: hailiu@asu.edu

**Education**

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|---------------------------------------------------------------------|-----------------------|
| <b>University of Southern California</b> , Los Angeles, CA, USA     | Aug. 2016 – Aug. 2022 |
| Ph.D. in Electrical Engineering                                     |                       |
| <b>Harbin Institute of Technology</b> , Harbin, Heilongjiang, China |                       |
| M.S. in Materials Science                                           | Sep. 2004 - July 2006 |
| B.S. in Metal Materials Engineering                                 | Sep. 2000 - July 2004 |

**Academic Experience**

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| <b>Assistant Professor</b>                                                                                                                                                                                                                                                | August 2026 – present |
| Electrical, Computer and Energy Engineering, Arizona State University, Tempe, Arizona, USA                                                                                                                                                                                |                       |
| Responsibilities:                                                                                                                                                                                                                                                         |                       |
| <ul style="list-style-type: none"><li>Research and teach advanced semiconductor electronics/photonics packaging and micro-electro-mechanical systems (MEMS), bridging device innovation with applications in computing, communication, sensing, and healthcare.</li></ul> |                       |

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|----------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Research Assistant (PhD student)</b>                                                                                    | May 2016 - July 2022 |
| Electrical and Computer Engineering, University of Southern California, Los Angeles, CA, USA.                              |                      |
| Advisor: Prof. Eun Sok Kim                                                                                                 |                      |
| Responsibilities:                                                                                                          |                      |
| <ul style="list-style-type: none"><li>Research and develop innovative MEMS sensors and actuators for healthcare.</li></ul> |                      |

**Key achievements**

- Designed, modeled, fabricated, and characterized novel piezoelectric MEMS resonant microphone arrays (RMAs) for lung sound monitoring, achieving high sensitivity and a signal-to-noise ratio (SNR) of 79–98 dBA. Integrated the RMAs with machine learning for lung sound classification, demonstrating improved detection of wheezing compared to conventional microphones and paving the way for wearable electronic stethoscopes. Awarded the Outstanding Student Paper Award (18/275) at IEEE MEMS 2020, Vancouver, Canada.
- Researched active noise cancellation (ANC) for hearing aids, developing two RMAs to separately capture speech and noise. Compared to traditional microphones, this approach achieved greater noise cancellation in the ANC, reducing noise levels by -20 to -25 dB, particularly in low-noise environments, due to the RMAs' high SNR.
- Researched novel MEMS ultrasound transducers for acoustic tweezers to trap and rotate particles in liquid for potential biomedical applications. The transducers were fabricated from lead zirconate titanate (PZT) with a patterned polydimethylsiloxane (PDMS) membrane, using soft lithography to achieve the desired acoustic performance.
- Developed two MEMS renewable energy generators: a pyroelectric lithium niobate device for wireless IC tamper detection, and a micro-fabricated coil-and-magnet device for non-resonant broadband vibration energy harvesting in IoT electronics, achieving a power density of up to 200.4  $\mu\text{W}/\text{cm}^3$ .

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## Industrial Experience

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**Hardware Engineering Technical Leader**

Feb 2025 – July 2026

Cisco Systems, Inc., Holmdel, NJ, USA

Responsibilities:

- Research and develop advanced microelectronics packaging for semiconductor electronic IC (EIC) and Si photonics IC (PIC)

Key achievements:

- Led development of 3D packaging integrating a Si photonics IC, electronic driver, and transimpedance amplifier for high-speed optical transceivers supporting AI data communication.
- Led development of optical engines for client optics and co-packaged optics (CPO), integrating PIC, EIC, and a Si switch within a single package for data center applications.

**Staff Package Assembly Engineer**

July 2022 – Feb 2025

InvenSense, Inc., San Jose, CA, USA

Responsibilities:

- Research and develop advanced microelectronics packaging for MEMS sensors and semiconductor application-specific integrated circuits (ASIC)

Key achievements:

- Led development of an ultra-thin MEMS ultrasonic fingerprint sensor for wearable electronics using wafer-level chip-scale packaging (WLCSP) with through-silicon vias (TSV), resulting in a miniaturized, low-cost, and highly reliable product.
- Led development of a bottom-port, open-cavity package for MEMS ultrasonic time-of-flight sensors used in ranging and presence sensing, resulting in a cost-effective, high-performance solution with strong manufacturability.
- Led development of high-density packaging for MEMS inertial measurement units (IMUs) used in motion sensing.
- Developed the course of WLCSP with MEMS and ASIC Integration.
- Streamlined package development procedures and documentation.

**Engineering Summer Intern**

May 2017 – August 2017

TE Connectivity Ltd., Fremont, CA, USA.

- Researched and integrated MEMS pressure sensors into invasive medical catheters for fractional flow reserve (FFR) measurement in coronary arteries.

**Semiconductor Packaging R&D Engineer**

July 2006 - July 2016

Samsung Semiconductor China R&amp;D Co. Ltd., Suzhou, Jiangsu, China

Responsibilities:

- Research and develop advanced semiconductor packaging for high-density memory, advanced processors, and power management integrated circuits.

Key achievements

- Developed a range of advanced semiconductor packaging products, including multi-chip package (MCP) for high-density memory, quad flat no-lead package (QFN) for power management ICs, flip-chip plastic ball grid array (FC PBGA) for TV processors, and chip-on-board (COB) for communication devices.
- Developed Cu wire bonding, offering lower cost and higher performance than traditional Au wire bonding, despite added process and reliability challenges, and led its first high-volume manufacturing (HVM) ramp-up at Samsung, earning the Samsung Electronics R&D Gold Prize.
- Researched the reliability of Cu and Au-Ag wire bonding, identifying key material and process factors affecting reliability in humid environments and exploring solutions for improvement. These

findings were published in top IEEE electronics packaging conferences (ECTC, EPTC) and have been widely cited.

- Researched solder joint reliability across various pad structures and finishes, including dent pads with Cu-OSP finish and flat pads with Ni-Au and various Ag finishes, for the advanced high-density FBGA package.
- Invented novel pad surface finishes, including a hybrid Ni/Au-and-OSP finish offering strong temperature cycle (TC) and drop reliability, and a low-cost Ni-Cu-OSP finish offering strong TC performance to replace the traditional, more expensive Ni/Au finish, resulting in U.S. Patents 9,504,152 B2 and 10,049,970 B2.
- Published 6 peer-reviewed papers and was granted 6 patents.

## Honors and awards

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|----------------------------------------------------------------------|-----------------------|
| • Outstanding Reviewer Award, Nature Microsystems & Nanoengineering. | Jan. 2024             |
| • Outstanding Student Paper Award Finalist at MEMS 2020 (18/275).    | Jan. 2020             |
| • Annenberg Fellowship at USC.                                       | Aug. 2016 – July 2020 |
| • Best Technical Lecturer Award at Samsung                           | 2013                  |
| • Samsung Electronics R&D Gold Prize                                 | 2009                  |

## Service

- Academic Journal reviewer
  - Springer Nature: Scientific Reports (2023, 2024), Microsystems and Nanoengineering (2023), Microsystem Technologies (2023, 2024)
  - ASME: Journal of Electronics Packaging (2023, 2024, 2025)
  - IEEE: Electronic Device Letter (2023, 2025), IEEE Transactions on Components, Packaging and Manufacturing Technology (2023, 2024), Sensors Journal (2025)
  - MDPI: Micromachines (2023), Sensors (2023, 2024, 2025), Remote Sensing (2023, 2024), Energies (2023), ENG (2023), Applied Science (2023)
  - IOP Publishing: Engineering Research Express (2024), Smart Materials and Structures (2024), Journal of Micromechanics and Microengineering (2025)
  - Other: AME Journal of Thoracic Disease (2023)
- Academic Conference Review Committee - IEEE Sensors 2023, 2024.
- IEEE Senior Members Application Review Panel (2025)

## Membership

- IEEE Senior Member

## Publications

### Semiconductor Packaging

- [1] Liu, Hai, Li Jin, Liqun Gu, Juanjuan Li, Ying Li, Qiang Chen, Maouha Du, Jianwei Zhou, and TaeSub Chang. "Solder joint reliability with various silver finish on PCB." In 2014 15th International Conference on Electronic Packaging Technology, pp. 396-400. IEEE, 2014.
- [2] Liu, Hai, Zhenqing Zhao, Qiang Chen, Jianwei Zhou, Maohua Du, Senyun Kim, Jonghyun Chae, and Myungkee Chung. "Reliability of copper wire bonding in humidity environment." In 2011 IEEE 13th Electronics Packaging Technology Conference, pp. 53-58. IEEE, 2011.
- [3] Chen, Qiang, Zhenqing Zhao, Hai Liu, Jonghyun Chae, Senyun Kim, and Myungkee Chung. "Investigation of various pad structure influence for copper wire bondability." In 2011 12th International Conference on Electronic Packaging Technology and High Density Packaging, pp. 1-4. IEEE, 2011.

- [4] Liu, Hai, Qi Chen, Zhenqing Zhao, Qian Wang, Jianfeng Zeng, and Jonghyun Chae. "Reliability of Au-Ag alloy wire bonding." In 2010 Proceedings 60th Electronic Components and Technology Conference (ECTC), pp. 234-239. IEEE, 2010.
- [5] Zhang, Jing, Hai Liu, and Jaisung Lee. "Board level cyclic bending test for MCP package." In 2007 9th Electronics Packaging Technology Conference, pp. 459-462. IEEE, 2007.
- [6] Liu, Hai, Jing Zhang, Song Chen, Maohua Du, Nufeng Feng, Qian Wang, and Taekoo Lee. "Board level thermal cycle reliability of BGA for a new type of pad structure with OSP surface finish." In 2007 8th International Conference on Electronic Packaging Technology, pp. 1-3. IEEE, 2007.

#### **MEMS Resonant Microphone Array (RMA)**

- [7] Sengupta, Anik, Akash Roy, Hongxiang Gao, Matin Barekatin, Hai Liu, and Eun Sok Kim. "Wearable Stethoscope Based on Resonant Microphone Array with Wireless Data Transfer." In 2024 IEEE 37th International Conference on Micro Electro Mechanical Systems (MEMS), pp. 101-104.
- [8] Liu, Hai, M. Barekatin, A. Roy, S. Liu, Y. Cao, Y. Tang, A. Shkel and E.S. Kim, "MEMS Piezoelectric Resonant Microphone Array for Lung Sound Classification," *Journal of Micromechanics and Microengineering*, 33, No. 4 (2023), 044003.
- [9] Dodson, A., A. Jostes, D. Cerrone, H. Liu, E. S. Kim, and R. M. Kato. "Electronic Stethoscope Lung Sounds Analysis Versus Spirometry to Detect Change With Bronchodilator Use in Pediatric Asthma Patients." In American Thoracic Society International Conference Meetings Abstracts, vol. 207, no. Supplement\_1, pp. A2951-A2951. Oxford University Press, 2023.
- [10] Liu, Hai, Song Liu, Anton A. Shkel, Yongkui Tang, and Eun Sok Kim. "Multi-band MEMS resonant microphone array for continuous lung-Sound monitoring and classification." In 2020 IEEE 33rd International Conference on Micro Electro Mechanical Systems (MEMS), pp. 857-860. IEEE, 2020. Outstanding Student Paper Award Finalist (18/275)
- [11] Liu, Hai, Song Liu, Anton A. Shkel, and Eun Sok Kim. "Active noise cancellation with MEMS resonant microphone array." *Journal of Microelectromechanical Systems* 29, no. 5 (2020): 839-845.
- [12] Liu, Hai, Song Liu, Anton A. Shkel, Yongkui Tang, and Eun Sok Kim. "MEMS resonant microphone array for lung sound classification." In 2019 IEEE International Electron Devices Meeting (IEDM), pp. 34-4. IEEE, 2019.

#### **MEMS Energy Generators**

- [13] Barekatin, Matin, Hai Liu, and Eun Sok Kim. "Wireless and Battery-Less Tamper Detection With Pyroelectric Energy Converter and High-Overtone Bulk Acoustic Resonator." *IEEE Sensors Journal* 22, no. 14 (2022): 14639-14646.
- [14] Fan, Shuyu, Yongkui Tang, Lurui Zhao, Hai Liu, Yufeng Wang, Dibo Hou, and Yunqi Cao. "Design optimization of microfabricated coils for volume-limited miniaturized broadband electromagnetic vibration energy harvester." *Energy Conversion and Management* 271 (2022): 116299.

#### **MEMS Ultrasonic Transducer**

- [15] Liu, Song, Pengcheng Sun, Mingyue Wang, Yujie Jiang, Jiaqi Li, Yuyu Jia, Zhenhuan Sun, Hai Liu, Haojian Lu, and Zuankai Wang. "Acousto-dewetting enables droplet microfluidics on superhydrophilic surfaces." *Nature Physics* (2025): 1-9.
- [16] Li, Jiaqi, Z. Sun, H. Liu and S. Liu, "MEMS Amplitude-Gradient Acoustic Transducer (AGAT) for Controllable Particle Transportation," 2025 IEEE International Ultrasonics Symposium (IUS), Utrecht, Netherlands, 2025, pp. 1-4, doi: 10.1109/IUS62464.2025.11201785.
- [17] Li, Jiaqi, Zhenhuan Sun, Hai Liu, and Song Liu. "Obstacle-crossing Self Focused Acoustic Transducer based on Bessel Beams." In 2024 IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium (UFFC-JS), pp. 1-4. IEEE, 2024.
- [18] Li, Jiaqi, Zhenhuan Sun, Yuyu Jia, Tang Li, Haojian Lu, Lurui Zhao, Hai Liu, and Song Liu. "MEMS First-Order Bessel Beam Acoustic Transducer for Particle Trapping And Controllable Rotating." In 2023 IEEE 36th International Conference on Micro Electro Mechanical Systems (MEMS), pp. 965-968. IEEE, 2023.

- [19] Jiaqi Li, Zhenhuan Sun, Hai Liu, Lurui Zhao, Teng Li, and Song Liu, "Bottle trap acoustic transducer with PDMS Lens for Particle Trapping", In 2023 IEEE International Ultrasonics Symposium (IUS), IEEE, 2023.
- [20] Sun, Zhenhuan, Jiaqi Li, Hai Liu, Lurui Zhao, Teng Li, and Song Liu. "Rotation-free high order Bessel beam acoustic transducer for particle trapping." In 2023 IEEE International Ultrasonics Symposium (IUS), IEEE, 2023.
- [21] Li, Jiaqi, Zhenhuan Sun, Hai Liu, Lurui Zhao, Teng Li, and Song Liu. "Twin Trap Acoustic Transducer with PDMS Fresnel Lens for Particle Trapping." In 2023 IEEE International Ultrasonics Symposium (IUS), IEEE, 2023.
- [22] Sun, Zhenhuan, Jiaqi Li, Hai Liu, Lurui Zhao, Teng Li, and Song Liu. "Ultrasonic Transducer encoding intersecting Bessel beams for particle patterning." In 2023 IEEE International Ultrasonics Symposium (IUS), IEEE, 2023.
- [23] Jia, Yuyu, Zhenhuan Sun, Jiaqi Li, Lurui Zhao, Hai Liu, Teng Li, and Song Liu. "MEMS Bessel Beam Acoustic Transducer (MEMS-BBAT) with Air-Cavity Lens Based on Spiral Diffraction Grating for Particle Trapping." In 2022 IEEE International Ultrasonics Symposium (IUS), 2022.
- [24] Sun, Zhenhuan, Jiaqi Li, Yuyu Jia, Yuhang Liu, Kunyong Lyu, Hai Liu, Hu Su, Teng Li, and Song Liu. "Particle Trapping and Transportation with MEMS Zeroth-order Bessel Beam Acoustic Transducer (BBAT)." In 2022 IEEE International Conference on Robotics and Biomimetics (ROBIO), pp. 1-6. IEEE, 2022.
- [25] Liu, Hai, A. Roy, Y. Tang, M. Barekatin and E.S. Kim, "Ultrasonic Air-Borne Propulsion Through Synthetic Jets," Solid-State Sensor and Actuator Workshop, Hilton Head Island, SC, June 5 - 9, 2022, pp. 226 - 229.

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### Granted Patents

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- [1] Liu, Hai. "Methods of manufacturing printed circuit board and semiconductor package." U.S. Patent 10,049,970 B2, issued August 14, 2018.
- [2] Liu, Hai. "Printed circuit board for semiconductor package." U.S. Patent 9,504,152 B2, issued November 22, 2016.
- [3] Liu, Hai. Surface Coating and Semiconductor Packaging Including the Surface Coating, Chinese patent 106954335 B, issued September 17, 2019.
- [4] Liu, Hai. Printed Circuit Board and Semiconductor Packaging Including It, Chinese patent 105307386 B, issued July 06, 2018.
- [5] Liu, Hai. A Mixed Surface Coating and Its Manufacturing Method, Chinese patent 103227160 B, issued March 16, 2016
- [6] Lu, Kai, Hai Liu. Mixed Solder Ball Arrangement and Its Formation Method, Chinese patent 103219310 B, issued July 13, 2016.

# Yong-Hang Zhang

## A. Professional Preparation:

|                                                          |                        |               |      |
|----------------------------------------------------------|------------------------|---------------|------|
| Nanjing Normal University                                | Physics                | B.S.          | 1982 |
| Institute of Semiconductors, Chinese Academy of Sciences | Device Physics         | M. Sc.        | 1987 |
| University of Stuttgart, Germany                         | Solid State Physics    | Dr. rer. nat. | 1991 |
| University of California at Santa Barbara                | Electrical Engineering | Postdoc       | 1993 |

## B. Appointments:

### Academic Appointments:

2005-present: Founding Director, Center for Photonics Innovation  
 1996 to present: Associate (96-00) and Professor (00-present), School of Electrical, Computer and Energy Engineering, ASU  
 2000 to present: Affiliated Professor in Physics and Materials Engineering  
 2016-2018: Director, ASU NanoFab  
 2014-2016: Associate Dean for Research, Ira A. Fulton Schools of Engineering  
 2012-2014: Fulton Entrepreneurial Professor, Ira A. Fulton Schools of Engineering  
 1991-1993: Assistant Research Engineer, NSF QUEST Center, UCSB

### Industrial Experience:

2012-2014: Founded Zipton Labs LLC and won two SBIR/STTR Phase I programs.  
 8/2000-5/2004: The Chairman of the Board, CEO, and President of Lytek Corporation, secured two round of equality financing and a venture band loan and won subcontracts of 1 SBIR Phase I and 1 SBIR Phase II program with total funding of \$7.3M.  
 5/2001-8/2003: On leave from ASU and the Chairman of the Board, CEO, and President of Lytek Corporation  
 5/2000: Founded Lytek Corporation  
 3/2000: Founded Optolocity LLC and won two SBIR programs  
 1995-1996: Research Staff Member, Hughes Research Laboratories  
 1993-1995: Postdoctoral Research Staff Member, Hughes Research Laboratories

## C. Publications:

### **Publications most closely related to the proposed project:**

1. Book chapter: Y.-H. Zhang, *Heterovalent semiconductor structures and their device applications*, in Molecular Beam Epitaxy, edited by Hajime ASahi and Yoshiji HORIKOSHI, John Wiley & Sons, Ltd, March, 2019.
2. Z. Ju, X. Qi, X. Liu, R. Hossain, A. Wang, T. McCarthy, A. McMin, Y.-H. Zhang, *Monocrystalline CdSeTe/MgCdTe Double-Heterostructure Solar Cells*, Solar RRL **9**(13), 2500138 (2025). (**Cover story**)
3. X. Qi, Z. Ju, X. Liu, J. Gong, Y. Lu, Y. Liu, R. Hossain, N. Rosenblatt, T. T. McCarthy, A. M. McMin, M. R. McCartney, D. J. Smith, Z. Ma, Y.-H. Zhang, *Interface Fermi-Level Engineering for Selective Hole Extraction Without p-Type Doping In CdTe Solar Cells to Reach High Open Circuit Voltage (>1 V)*, Solar RRL **9**(12), 2500124 (2025). (**Cover story**).
4. F. Zhang, J. F. Castaneda, T. H. Gfroerer, D. Friedman, Y.-H. Zhang, M. W. Wanlass, and Y. Zhang, *An all optical approach for comprehensive in-operando analysis of radiative and nonradiative*

*recombination processes in semiconductors*, SPRINGER NATURE Journal Light: Science & Applications **11**, 137 (2022)

5. Y. Zhou, Y. Miao, S. Ojo, H. Tran, G. Abernathy, J. M Grant, S. Amoah, G. Salamo, W. Du, J. Liu, J. Margetis, J. Tolle, Y.-H. Zhang, G. Sun, R. A. Soref, B. Li, S.-Q. Yu, *Electrically injected GeSn lasers on Si operating up to 100 K*, Optica **7**, 924-928 (2020).
6. Y. Zhao, M. Boccard, S. Liu, J. Becker, X.-H. Zhao, C. M. Campbell, E. Suarez, M. B. Lassise, Z. Holman, and Y.-H. Zhang, *Monocrystalline CdTe Solar Cells with Open-Circuit Voltage Over 1V and Efficiency of 17%*, Nature Energy **1**, 16067 (2016).
7. E. H. Steenbergen, B. C. Connelly, G. D. Metcalfe, H. Shen, M. Wraback, D. Lubyshev, Y. Qiu, J. M. Fastenau, A. W. K. Liu, S. Elhamri, O. O. Cellek, and Y.-H. Zhang, *Significantly improved minority carrier lifetime observed in a long-wavelength infrared III-V type-II superlattice comprised of InAs/InAsSb*, Appl. Phys. Lett. **99**, 251110 (2011)
8. H. S. Kim, O. O. Cellek, Z. Lin, Z.-Y. He, H. Li, S. Liu, and Y.-H. Zhang, *Long-wave infrared nBn photodetectors based on InAs/InAsSb type-II superlattices*, Appl. Phys. Lett. **101**, 161114 (2012).
9. Y.-H. Zhang, *Continuous Wave Operation of InAs/InAs<sub>x</sub>Sb<sub>1-x</sub> Mid Infrared Lasers*, Appl. Phys. Lett. **66**, 118 (1995).
10. Y.-H. Zhang and D.H. Chow, *Improved Crystalline Quality of AlAs<sub>x</sub>Sb<sub>1-x</sub> Grown on InAs by Modulated Molecular-Beam Epitaxy*, Appl. Phys. Lett. **65**, 3239 (1994)

#### Other significant publications:

11. J. Ding, C.-Y. Tsai, Z. Ju, and Y.-H. Zhang, *Epitaxial lift-off CdTe/MgCdTe double heterostructures for thin-film and flexible solar cells applications*, Appl. Phys. Lett. **118**, 181101 (2021). (**Featured article, Editor's pick as Scilight**)
12. C.-Y. Tsai, Y. Zhang, Z. Ju, and Y.-H. Zhang, *Study of vertical hole transport in InAs/InAsSb type-II superlattices by steady-state and time-resolved photoluminescence spectroscopy*, Appl. Phys. Lett. **116**, 201108 (2020)
13. E. Luna, A. Trampert, J. Lu, T. Aoki, Y.-H. Zhang, M.R. McCartney, D.J. Smith, *Strategies for Analyzing Noncommon-Atom Heterovalent Interfaces: The Case of CdTe-on-InSb*, Advanced Materials Interfaces, 1901658 (2019).
14. M.B. Lassise, T.T. McCarthy, B.D. Tracy, D.J. Smith, Y.-H. Zhang, *Molecular beam epitaxial growth and structural properties of hetero-crystalline and heterovalent PbTe/CdTe/InSb structures*, J. of Applied Physics **126** (4), 045708 (2019).
15. X. Wang, C. M. Campbell, Y.-H. Zhang, R. J. Nemanich, *Band alignment at the CdTe/InSb (001) heterointerface*, J. Vac. Sci. Technol. A **36**, 031101-5 (2018) (**Editor's pick**)
16. J.-J. Li, L. Yin, S. R Johnson, B. J. Skromme, S. Wang, X. Liu, D. Ding, C.-Z. Ning, J. K. Furdyna, Y.-H. Zhang, *Photoluminescence studies of type-II CdSe/CdTe superlattices*, Appl. Phys. Lett. **101**, 061915-061915 (2012).

#### D. Synergistic Activities:

1. AI Cho MBE Award by the 24th International Conference on MBE, 2026
2. Art Gossard MBE Innovator Award by the 39th North American Conference on MBE, 2025.
3. IEEE and OSA Fellow
4. PI or Co-PI of 4 DOD MURI, Co-PI of 1 DOE EFRC and 1 NSF ERC programs since 2005.
5. Co-General Chair for the 14<sup>th</sup> Int. Conf. on Mid-IR Optoelectronics: Materials and Devices, 2018
6. General chair of the 18<sup>th</sup> Int. MBE Conference, 2014
7. The chair of both Advisory Committees for the International Conference on Molecular Beam Epitaxy (MBE) and the International North America MBE Conference
8. He has graduated 33 PhD students and supervised over 45 postdocs and visiting scholars.

## Curriculum Vitae

# Xiangfan Chen

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Group Website: <https://faculty.engineering.asu.edu/chen/>  
Phone: +1-(480)-727-2662

### EDUCATION

Ph.D. 06/2018      **Northwestern University**, Mechanical Engineering, Evanston, IL, USA  
**Thesis:** “Scalable Micro and Nano Manufacturing of Photonic Devices”  
**Advisor:** Dr. Cheng Sun  
B.S. 06/2012      **Shanghai Jiao Tong University**, Mechanical Engineering, China  
**Thesis:** “Design and Fabrication of Flapping-Wing Micro Air Vehicle”  
**Advisor:** Dr. Weiping Zhang

### RESEARCH AREA & INTERESTS

**Manufacturing:** Multi-Material and Multi-Scale Three-dimensional (3D) Printing;  
**Printable Materials:** Hydrogels, Ceramics, Metals, and Composites;  
**Functional Devices:** Advanced Sensing, Soft Robotics, Free-form Optics, Batteries, Tissue Engineering, Regenerative Medicine;  
**Metamaterials:** Photonic/ Phononic/ Mechanical Metamaterials.

### PROFESSIONAL EXPERIENCE

08/2026 – Present      **Associate Professor** in Manufacturing Engineering, School of Manufacturing Systems and Networks (MSN), Ira A. Fulton Schools of Engineering (FSE), Arizona State University (ASU)  
08/2018 – 07/2026      **Assistant Professor** in Manufacturing Engineering, System Engineering, MSN, FSE, ASU  
                                 **Graduate Faculty** in Materials Science and Engineering, Mechanical Engineering, School for Engineering of Matter, Transport & Energy (SEMTE), FSE, ASU  
                                 **Faculty**, The Biomimicry Center, Center for Complex System Safety  
09/2012 – 06/2018      **Graduate Research Assistant**, Department of Mechanical Engineering, Northwestern University

### HONORS & AWARDS

2025      Arizona Biomedical Research Commission - New Investigator Award (ABRC-NIA)  
2025      1<sup>st</sup> Place Poster Award, The Minerals, Metals & Materials Society (TMS) 2025  
2024      Susan Smyth Outstanding Young Manufacturing Engineers Award, The Society of Manufacturing Engineers (SME)  
2022      Mayo Clinic-ASU Alliance Fellow in the 2022 Mayo Clinic and ASU Alliance for Healthcare Summer Residency Program  
2016      E. Cabell Terminal Year Fellowship, Northwestern University  
2015      Martin Outstanding Graduate Fellowship, Northwestern University  
2012      Walter P. Murphy Fellowship, Northwestern University

## PUBLICATIONS

Google Scholar: <https://scholar.google.com/citations?user=Kzlc974AAAAJ&hl=en>

**Journal Publications** (\* Corresponding Author, <sup>1</sup> Co-First Author)

### At ASU

#### **Published & Accepted**

1. Liu, L., Kublik, N., Wang, W., Niauzorau, S., Zhu, Y., Carvalho, F., Wang, Y., Liu, S., Jayachandran, A., Azeredo, B.\*, & **Chen, X.\*** (2025). Rapid 3D Printing of Hierarchical Nanoporous Copper Structures with Self-Disintegration Capability. *Nature Communications*, 17, 695.
2. Liu, L., Wang, W., Rajeev, S., Wang, Y., Layazali, S., Tong, T., Ma, C., & **Chen, X.\*** (2025). 3D Printing of Multiscale Structured Aerogels for Efficient Solar Steam Generation. *Advanced Energy and Sustainability Research*, 000e202500293.
3. Wang, W., Liu, S., Liu, L., Ma, X., Nogueira, C., Shefi, O., Lancaster, J., & **Chen, X.\*** (2025). Dynamic fluid-assisted continuous multi-material 3D printing for seamless gradient structures. *Advanced Material Technologies*, 2500621.
4. Subedi, S.<sup>1</sup>, Liu, S.<sup>1</sup>, Wang, W., Shavon, A., **Chen, X.\***, & Ware, H.\* (2024). Multi-Material Vat Photopolymerization 3D Printing: A Review of Mechanisms and Applications. *npj Advanced Manufacturing*, 1, 9.
5. Wang, W., Liu, S., Yu, M., **Chen, X.\*** (2024). Rapid 3D printing of electro-active hydrogels. *Manufacturing Letters*, 41, 862-867.
6. Zhu, Y., Liu, S., Mei, X., Lin, Z., Pulido, T. V., Hou, J., Remani, S. A., Patil, D., Sobczak, M. T., Ramanathan, A., Thummalapalli, S. V., Chambers, L. B., Yu, C., Guo, S., Zhao, Y., Liu, Y., Wang, X., Lancaster, J. N., Zhang, Y. S., **Chen, X.**, & Song, K. (2024). 3D Coaxial Printing of Small-Diameter Artificial Arteries. *Small Structures*, 2400323.
7. Patil, D., Liu, S., Ravichandran, D., Thummalapalli, S. V., Zhu, Y., Tang, T., Golan, Y., Miquelard-Garnier, G., Asadi, A., Li, X., **Chen, X.**, & Song, K. (2024). Versatile Patterning of Liquid Metal via Multiphase 3D Printing. *Small*, 2402432.
8. Liu, L., Wang, W., Feng, S., Liu, S., Sun, H., Nian, Q., Yang, S., & **Chen, X.\*** (2024). Rapid, Micron-Resolution 3D Printing of Nd:YAG Ceramic with Optical Gain. *Small*, 2403130.
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11. Liu, S., Alfarhan, S., Wang, W., Feng, S., Zhu, Y., Liu, L., Song, K., Yang, S., Jin, K., & **Chen, X.\*** (2023). Robust Optical Data Encryption by Projection-Photoaligned Polymer-Stabilized-Liquid-Crystals. *Advanced Optical Materials*, 11(21), 2301364. **[Back Inside Cover]**
12. Wang, W., Liu, S., Liu, L., Alfarhan, S., Jin, K., & **Chen, X.\*** (2023). High-Speed and High-Resolution 3D Printing of Self-Healing and Ion-Conductive Hydrogels via  $\mu$ CLIP. *ACS Materials Letters*, 5, 1727-1737.
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14. Zhu, Y., Kwok, T., Haug, J. C., Guo, S., **Chen, X.**, Xu, W., Ravichandran, D., Tchoukalova, Y.D., Cornella, J.L., Yi, J. and Shefi, O., Vernon, B., Lott, D., Lancaster, J., & Song, K. (2023).

- 3D Printable Hydrogel with Tunable Degradability and Mechanical Properties as a Tissue Scaffold for Pelvic Organ Prolapse Treatment. *Advanced Materials Technologies*, 8(10), 2201421.
15. Liu, S., Wang, W., Xu, W., Liu, L., Zhang, W., Song, K., & **Chen, X.\*** (2022). Continuous three-dimensional printing of architected piezoelectric sensors in minutes. *Research, A Science Partner Journal*, 2022, 9790307.
  16. Xu, W., Jambhulkar, S., Ravichandran, D., Zhu, Y., Kakarla, M., Nian, Q., Azeredo, B., **Chen, X.**, Jin, K., Vernon, B., Lott, D.G., Cornella, J., Shefi, O., Miquelard-Garnier, G., Yang, Y., & Song, K. (2021). 3D printing-enabled nanoparticle alignment: a review of mechanisms and applications. *Small*, 17(45), 2100817.
  17. Jambhulkar, S., Liu, S., Vala, P., Xu, W., Ravichandran, D., Zhu, Y., Bi, K., Nian, Q., **Chen, X.**, & Song, K. (2021). Aligned Ti3C2T x MXene for 3D Micropatterning via Additive Manufacturing. *ACS Nano*, 15(7), 12057-12068.
  18. Liu, L., Liu, S., Schelp, M., & **Chen, X.\*** (2021). Rapid 3D printing of bioinspired hybrid structures for high-efficiency fog collection and water transportation. *ACS Applied Materials & Interfaces*, 13(24), 29122-29129.

#### **Before ASU**

19. Huang, J., **Chen, X.**, Bai, P., Hai, R., Sun, C., & Xu, T. (2019). 45% Periodicity Reduction in Nanocomposite Thin Films via Rapid Solvent Removal. *Macromolecules*, 52(4), 1803-1809.
20. **Chen, X.**, Dong, B., Wang, C., Zhou, F., & Sun, C. (2018). Hyperbolic Dispersion via Symmetric and Antisymmetric Orderings of Artificial Magnetic Dipole Array. *ACS Photonics*, 5(11), 4469-4475.
21. Li, H., Dong, B., Zhang, X., Shu, X., **Chen, X.**, Hai, R., Czaplewski, D.A., Zhang, H.F., & Sun, C. (2019). Disposable ultrasound-sensing chronic cranial window by soft nanoimprinting lithography. *Nature communications*, 10(1), 4277.
22. **Chen, X.**, Liu, W., Dong, B., Lee, J., Ware, H. O. T., Zhang, H. F., & Sun, C. (2018). High-speed 3D printing of millimeter-size customized aspheric imaging lenses with sub 7 nm surface roughness. *Advanced Materials*, 30(18), 1705683. **[Front Cover] (Featured on Science Daily, Phys.org, EurekAlert by AAAS, McCormick News, etc.)**
23. **Chen, X.**, Ware, H. O. T., Baker, E., Chu, W., Hu, J., & Sun, C. (2017). The development of an all-polymer-based piezoelectric photocurable resin for additive manufacturing. *Procedia CIRP*, 65, 157-162.
24. Ware, H. O. T., Farsheed, A. C., Akar, B., Duan, C., **Chen, X.**, Ameer, G., & Sun, C. (2018). High-speed on-demand 3D printed bioresorbable vascular scaffolds. *Materials Today Chemistry*, 7, 25-34.
25. Dong, B., Soetikno, B. T., **Chen, X.**, Backman, V., Sun, C., & Zhang, H. F. (2017). Parallel three-dimensional tracking of quantum rods using polarization-sensitive spectroscopic photon localization microscopy. *ACS Photonics*, 4(7), 1747-1752.
26. Yu, S., Wang, C., Zhang, Y., Dong, B., Jiang, Z., **Chen, X.**, Chen, W., & Sun, C. (2017). Design of non-deterministic quasi-random nanophotonic structures using Fourier space representations. *Scientific Reports*, 7(1), 3752.
27. Dong, B.<sup>1</sup>, **Chen, X.**<sup>1</sup>, Zhou, F., Wang, C., Zhang, H. F., & Sun, C. (2016). Gigahertz all-optical modulation using reconfigurable nanophotonic metamolecules. *Nano Letters*, 16(12), 7690-7695.
28. **Chen, X.**<sup>1</sup>, Guo, P.<sup>1</sup>, He, C., Dong, B., Ocola, L. E., Schaller, R. D., Chang, R.P., & Sun, C.

- (2016). Scaling the Artificial Polariton Bandgap at Infrared Frequencies Using Indium Tin Oxide Nanorod Arrays. *Advanced Optical Materials*, 4(12), 2077-2084.
29. Zhu, W. L., Zhu, Z., Shi, Y., Chen, X., He, Y., Ehmann, K. F., & Ju, B. F. (2016). A novel piezoelectrically actuated 2-DoF compliant micro/nano-positioning stage with multi-level amplification. *Review of Scientific Instruments*, 87(10).
  30. Zhou, F., Wang, C., Dong, B., Chen, X., Zhang, Z., & Sun, C. (2016). Scalable nanofabrication of U-shaped nanowire resonators with tunable optical magnetism. *Optics Express*, 24(6), 6367-6380.
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### **Peer-Reviewed Proceedings**

32. Liu, L., Kublik, N., Azeredo, B., & Chen, X.\* (2023, June). Rapid 3D Printing of Nanoporous Copper Powders via Micro-Clip. In International Manufacturing Science and Engineering Conference (Vol. 87233, p. V001T01A016). *American Society of Mechanical Engineers*.
33. Shao, G., Ware, H. O. T., Chen, X., Li, L., & Sun, C. (2019, March). High-resolution 3D printing magnetically-active microstructures using micro-CLIP process. In Nano-, Bio-, Info-Tech Sensors and 3D Systems III (Vol. 10969, pp. 69-74). *SPIE*.
34. Chen, X., Liu, W., Dong, B., Ware, H. O. T., Zhang, H. F., & Sun, C. (2018, June). High-throughput 3D printing of customized imaging lenses. In International Manufacturing Science and Engineering Conference (Vol. 51357, p. V001T01A023). *American Society of Mechanical Engineers*.
35. Chen, X., Guo, P., Dong, B., Zhang, Z., Chang, R. P., & Sun, C. (2016, October). Theoretical and experimental manipulation of plasmon-polariton bandgaps at infrared frequencies in indium-tin-oxide nanorod arrays. In 2016 IEEE Photonics Conference (IPC) (pp. 637-638). *IEEE*.
36. Wang, C., Yu, S., Dong, B., Zhang, Y., Jiang, Z., Chen, X., Zi, J., Chen, W., & Sun, C. (2016, June). Bio-inspired design strategy of quasi-random structures for optimal light control. In CLEO: Applications and Technology (pp. AW1K-5). *Optica Publishing Group*.
37. Chen, X., Wang, C., Baker, E., Wang, J., & Sun, C. (2014, March). Understanding the nanophotonic light-trapping structure of diatom frustule for enhanced solar energy conversion: a theoretical and experimental study. In Bioinspired, Biointegrated, Bioengineered Photonic Devices II (Vol. 8958, pp. 23-32). *SPIE*.

## **PRESENTATIONS**

### **Invited Talks and Seminars**

1. “Development of New Multi-Material Additive Manufacturing Platform for Functional Materials”. Seminar, Edwardson School of Industrial Engineering (IE), Purdue University, November 2024.
2. “High-Speed and High-Resolution Additive Manufacturing of Functional Hydrogels”. Seminar, Biodesign Institute, Center for Biomaterials Innovation and Translation, Arizona State University, October 2024.
3. “Rapid 3D Printing of Functional Devices for Biomedical Applications via Micro-CLIP”. Invited Talk, Mayo Clinic, Scottsdale, AZ, August 2022.
4. Seminar (Zoom), Department of Mechanical Engineering, McGill University, October 2021.

5. Seminar, Center of Surface Engineering and Tribology, Northwestern University, Evanston, IL, May 2018.
6. Seminar, The Polytechnic School, Ira A. Fulton Schools of Engineering, Arizona State University, Mesa, AZ, April 2018.
7. Seminar, The Department of Mechanical Engineering, Northern Illinois University, DeKalb, IL, February 2018.
8. Seminar, The School of Mechanical and Aerospace Engineering, Oklahoma State University, Stillwater, OK, January 2018.

### **Conference Presentations**

9. [Invited Talk] Xiangfan Chen\*, Micro Additive Manufacturing of Functional Hydrogels. TMS 2025 Annual Meeting & Exhibition, March 23 – 27, 2025, Las Vegas, NV, United States.
10. [Invited Talk] Xiangfan Chen\*, High-Speed and High-resolution 3D Printing of Neodymium-doped Yttrium Aluminum Garnet Ceramics with Optical Gain Properties. The ACerS Electronic Materials and Applications (EMA) 2024, February 13 – 16, 2024, Denver, CO, United States.
11. Siying Liu, Xiangfan Chen\*, High Resolution Three-Dimensional Printing of Piezoelectric Composites for Sensing. International Conference on Advanced Manufacturing (ICAM) 2023, October 30 -November 03, 2023, Washington, DC, United States.
12. Xiangfan Chen\*, Wenbo Wang, High Speed and High-Resolution 3D Printing of Self-Healing and Ion-Conductive Hydrogels via  $\mu$ CLIP. International Conference on Advanced Manufacturing (ICAM) 2023, October 30 -November 03, 2023, Washington, DC, United States.
13. Luyang Liu, Xiangfan Chen\*, Rapid 3D Printing of Nanoporous Copper Powders via Micro-Clip. 2023 International Manufacturing Science and Engineering Conference (MSEC2023), June 12 – 16, 2023, New Brunswick, NJ, United States.
14. Luyang Liu, Xiangfan Chen\*, Rapid 3D Printing of AlN Ceramic for Heat Dissipation Devices. TMS 2023 Annual Meeting & Exhibition, March 19 – 23, 2023, San Diego, CA, United States.
15. Wenbo Wang, Siying Liu, Luyang Liu, Xiangfan Chen\*, High-Speed and High-Resolution 3D Printing of Self-healing and Ion-Conductive Hydrogels via  $\mu$ CLIP. TMS 2023 Annual Meeting & Exhibition, March 19 – 23, 2023, San Diego, CA, United States.
16. Luyang Liu, Xiangfan Chen\*, Rapid 3D Printing of Nd:YAG Ceramic for Lasing. TMS 2023 Annual Meeting & Exhibition, March 19 – 23, 2023, San Diego, CA, United States.
17. Xiangfan Chen\*, (Invited), Rapid Three-dimensional Printing of High-resolution Piezoelectric Devices. MS&T22: Materials Science & Technology, October 9 – 12, 2022, Pittsburgh, PA, United States.
18. Siying Liu, Xiangfan Chen\*, Three-dimensional Printing of High-Resolution Piezoelectric Structures. TechConnect World Innovation, June 13 – 15, 2022, Washington, DC, United States.
19. Siying Liu, Wenbo Wang, Luyang Liu, Xiangfan Chen\*, Rapid Three-dimensional Printing of High-resolution Piezoelectric Structures Using Micro-CLIP. SPIE Smart Structures + NDE, March 06 – 10, 2022, Long Beach, CA, United States.
20. Wenbo Wang, Siying Liu, Xiangfan Chen\*, 3D Printing of Ion-Conductive and Self-Healing Hydrogels for Customized Wearable Sensors. SPIE Smart Structures + NDE, March 06 – 10, 2022, Long Beach, CA, United States.
21. Siying Liu, Xiangfan Chen\*, “High-resolution 3D Printing of Piezoelectric Structures Using

- Micro-CLIP Process”, 2021 Virtual Material Research Society (MRS) Spring Meeting, April 17 – 23, 2021.
22. Xiangfan Chen\*, “Hyperbolic Dispersion via Symmetric and Antisymmetric Orderings of Artificial Magnetic Dipole Array”, 2019 SPIE Optics + Photonics, August 11 – 15, 2019, San Diego, CA, United States.
  23. Rishabh Kabade, Xiangfan Chen\*, “High-speed and high-resolution 3D printing of customized photonic devices”, 2019 SPIE Optics + Photonics, August 11 – 15, 2019, San Diego, CA, United States.
  24. Xiangfan Chen, “High-Throughput 3D Printing of Customized Imaging Lenses”, 2018 ASME Manufacturing Science and Engineering Conference (MSEC), June 18 – 22, 2018, College Station, TX.
  25. Xiangfan Chen, “High-Throughput 3D Printing of Millimeter-Size Customized Aspheric Imaging Lenses”, Material Research Society (MRS) Spring Meeting, Apr. 02 – 06, 2018, Phoenix, AZ.
  26. Xiangfan Chen, “Gigahertz All-Optical Modulation Using Reconfigurable Plasmonic Metamaterials”, Material Research Society (MRS) Spring Meeting, Apr. 02 – 06, 2018, Phoenix, AZ.
  27. Xiangfan Chen, Wenzhong Liu, Biqin Dong, Hao F. Zhang, Cheng Sun, “High-Throughput 3D Printing of Customized Imaging Lenses”, SPIE Photonics West, Jan. 27 – Feb. 01, 2018, San Francisco, CA.
  28. Xiangfan Chen, “The Development of an All-Polymer-Based Piezoelectric Photocurable Resin for Additive Manufacturing”, 3rd CIRP Conference on BioManufacturing, July 11 – 14, 2017, Chicago, IL.
  29. Xiangfan Chen, “Gigahertz All-Optical Modulation Using Reconfigurable Plasmonic Metamolecules”, 59th Electronic Materials Conference (EMC), June 28 – 30, 2017, South Bend, IN.
  30. Xiangfan Chen, “Theoretical and Experimental Manipulation of Plasmon-Polariton Bandgaps at Infrared Frequencies in Indium-Tin-Oxide Nanorod Arrays”, 2016 IEEE Photonics Conference (IPC), Oct. 02 – 06, 2016, Waikoloa, HI.
  31. Xiangfan Chen, “Free-Standing U-Cavity Array with Strong and Tunable Artificial Magnetism at Optical Frequencies”, Material Research Society (MRS) Fall Meeting, Dec. 01 – 05, 2014, Boston, MA.
  32. Xiangfan Chen, “Understanding the Nanophotonic Light-Trapping Structure of Diatom Frustule for Enhanced Solar Energy Conversion: A Theoretical and Experimental Study”, SPIE Photonics West, Feb. 01 – 07, 2014, San Francisco, CA.

## **PATENT**

1. Xiangfan Chen, Wenbo Wang, “A Dynamic Fluid-Enabled Multi-Material 3D Printing Platform”, Provisional application #63/733,722.
2. Xiangfan Chen, Luyang Liu, and Chao Ma, “Rapid 3D printing of aerogels with bioinspired surface structures for solar steam generation”, Provisional application #63/727,873.
3. Xiangfan Chen, Wenzhong Liu, Hao F. Zhang, and Cheng Sun, “High-throughput 3D Printing of Customized Aspheric Imaging Lenses”, US Patent Application, US20210016496A1.
4. Siying Liu, Wenbo Wang, and Xiangfan Chen, “Continuous three-dimensional printing of architected piezoelectric sensors”, US Patent Application, US20230303853A1.
5. Yash Mistry, Siying Liu, Mandar Shinde, Xiangfan Chen, Dhruv Bhate, Nikhilesh Chawla,

Swapnil Morankar, and Clint Penick, “Interwoven lattice structure and method for designing ultra-compliant interwoven meta-materials”, US20240093479A1.

## **STUDENT MENTORING & THESIS ADVISING**

### **Ph.D. Students Mentored: [5]**

#### *Current Ph.D. Students: [2]*

1. Yitian Wu, Manufacturing Engineering, MSN, Summer 2026 – present, Chair
2. Yao Wang, Manufacturing Engineering, MSN, Fall 2024 – present, Chair
3. Negar Akrami, Manufacturing Engineering, MSN, Spring 2025 – present, Chair.

#### *Graduated Ph.D. Students: [3]*

1. Wenbo Wang, Manufacturing Engineering, MSN, Fall 2020 – Spring, Chair. **Dissertation:** Dynamic-Fluid-Assisted Micro Continuous Liquid Interface Production of Functional Graded Materials. [Current position: TBD]
2. Luyang Liu, Manufacturing Engineering, MSN, Spring 2020 – Spring 2025, Chair. **Dissertation:** Scalable Continuous 3D Printing of Polymer Nanocomposites: Innovations for Functional Applications. [Current position: TBD]
3. Siying Liu, Materials Science and Engineering, SEMTE, Spring 2020 – Summer 2024, Chair. **Dissertation:** Light-based Advanced Manufacturing and Manipulation of Functional Polymers and Composites. [Current position: Research Scientist, BoronMatrix. Inc.]

### **M.S. Students Mentored: [10]**

#### *Current M.S. Students: [3]*

1. Pratyodhaya Padalinathan, Mechanical Engineering, SEMTE, Spring 2025 – present, Thesis Chair
2. Pankaj Mishra, Manufacturing Engineering, Fall 2024 – present, Thesis Chair.

#### *Graduated M.S. Students: [7]*

3. Sonia Hernandez, Robotics, MSN, Spring 2021 – Spring 2025, Thesis Chair. **Thesis:** 3D Printing of Electroactive Hydrogels Enhanced with Reduced Graphene Oxide (rGO). [Current position: Deloitte]
4. Chayaank Ravishankar, Manufacturing Engineering, MSN, Fall 2022 – Spring 2024, Thesis Chair. **Thesis:** Rapid 3D Printing of Reduced Graphene Oxide/Polymer Composites Using Micro-Continuous Liquid Interface Production. [Current position: Applied Materials]
5. Ujjawal Jha, Mechanical Engineering, SEMTE, Fall 2022 – Fall 2023, Thesis Chair. **Thesis:** Explore Rapid 3D Printing of Magnetically Actuated Microstructures Via Micro-Continuous Liquid Interface Production. [Current position: Micron Technology]
6. Jacob Yoshitake, Mechanical Engineering System, TPS, Fall 2021 – Fall 2022, Applied Project Chair.
7. Rishabh Kabade, Mechanical Engineering, SEMTE, Fall 2018 – Spring 2020, Applied Project Chair. [Current position: Eminence Equipments Pvt. Ltd]
8. Asjad Malik, Mechanical Engineering, SEMTE, Fall 2020 – Fall 2021, Research Project Mentor. [Current position: Allegro MicroSystems]
9. Chintan Shah, Mechanical Engineering, SEMTE, Fall 2019 – Fall 2020, Research Project Mentor. [Current position: Neptec optical Solutions]

**B.S. Students Mentored: [12]****Barrett Honors Thesis:**

1. Tiffany V. Pulido, Bioengineering and Biomedical Engineering, Barrett, The Honors College, Fall 2023 – Spring 2024, Barrett Honors Thesis Chair.
2. Alexander Potts, Robotics, Barrett, The Honors College, Fall 2022 – Spring 2023, Barrett Honors Thesis Co-Chair.
3. Daniel Mester, Mechanical System Engineering, Barrett, The Honors College, Fall 2020 – Spring 2021, Barrett Honors Thesis Chair.

**Others:**

4. Arick Nitzsche, Mechanical Engineering, Summer 2024 – Present, FURI, Undergraduate Research Volunteer.
5. Timothy Ong, Mechanical Engineering, The Honors College, Fall 2023 – Present, FURI, Undergraduate Research Volunteer.
6. Harrison LaBell, Mechanical System Engineering, Barrett, The Honors College, Fall 2023 – Fall 2024, Undergraduate Research Volunteer.
7. Riana Chatterjee, Computer Science, Barrett, The Honors College, Fall 2023 – Present, Undergraduate Research Volunteer.
8. Maxime Zand, Mechanical Engineering, Barrett, The Honors College, Spring 2023 – Present, Undergraduate Research Volunteer.
9. Sonia Hernandez, Robotics, TPS, Spring 2019 – Fall 2020, FURI, Undergraduate Research Volunteer.
10. Marcus Dopilka, Robotics, TPS, Fall 2021, Undergraduate Research Volunteer.
11. Michael Schelp, Robotics, TPS, Fall 2019 – Spring 2021, FURI, Undergraduate Research Volunteer.
12. Pradyumna Mysore, Mechanical Engineering, SEMTE, Spring 2019 – Summer 2020, FURI. Undergraduate Research Volunteer.

**High School Students Mentored: [8]**

1. Adwita Goyal, 2024 – 2025.
2. Nivid Singhanian, 2024 – 2025.
3. Phil Ni, 2024 – 2025.
4. Hannah Kim, 2023 – 2024.
5. Abhiram Kilaru, 2023 – 2024.
6. Karthik Raman, 2023.
7. Lekha Balakrishna, 2023.
8. Srishti Rajeev, 2022 – 2023.

**Dissertation Committee Member:**

1. Itzel Chavez Martinez, Ph.D. in Manufacturing Engineering, MSN, Thesis, Committee Member.
2. Zhaohong Xiu, Ph.D. in Manufacturing Engineering, MSN, Thesis, Committee Member.
3. Hongji Zhang, Ph.D. in Manufacturing Engineering, MSN, Thesis, Committee Member.
4. Natalya Kublik, Ph.D. in Materials Science and Engineering, SEMTE, Spring 2025, Thesis, Committee Member.
5. Saleh Alfarhan, Ph.D. in Chemical Engineering, SEMTE, Spring 2024, Comprehensive Exam, Committee Member.

6. Gabriel Nile, Ph.D. in Chemical Engineering, SEMTE, Comprehensive Exam (Summer 2023), Dissertation Defense (Spring 2024), Committee Member.
7. Weiheng Xu, Ph.D. in Systems Engineering, TPS, Summer 2022, Thesis, Committee Member.
8. Faisal Riyad, Ph.D. in Mechanical Engineering, SEMTE, Spring 2022, Thesis Comprehensive Exam, Committee Member.
9. Aliaksandr Sharstniou, Ph.D. in Materials Science and Engineering, SEMTE, Fall 2022, Thesis, Committee Member.

## **PROFESSIONAL SERVICE & ACADEMIC REVIEW**

**Grant Reviewer:** National Science Foundation (NSF), Arizona Biomedical Research Center (ABRC), ASU-Mayo Seed Grant.

**Editor Board Member:** npj Advanced Manufacturing.

**Guest/Topic Editor:** International Journal of Bioprinting, JoVE: Methods Collections, Journal of Micromachines.

**Paper Reviewer:** Advanced Materials, Small, Nano Energy, Light: Science & Applications by Nature Portfolio, Scientific Reports, Advanced Materials Interfaces, Advanced Materials Technologies, Additive Manufacturing, Journal of Manufacturing Processes, Optics Letters, Optics Express, Photonics Research, Applied Physics A, ASME Manufacturing Science and Engineering Conference (MSEC).

**Conference Session Organizer:** ASME Aerospace Structures, Structural Dynamics, and Materials Conference (SSDM 2023), Advanced Manufacturing Session.

**Conference Session Chair:** ASME Aerospace Structures, Structural Dynamics, and Materials Conference (SSDM 2023); MS&T22: Materials Science & Technology 2022; Advances in Micro- and Nano-Additive Manufacturing, Manufacturing Science and Engineering Conference (MSEC) 2023, 2018.

## **PROFESSIONAL SOCIETIES**

Member of Society of Manufacturing Engineer (SME), The Minerals, Metals & Materials Society (TMS), The American Ceramic Society (ACerS), American Institute of Chemical Engineers (AIChE), The Materials Research Society (MRS)

## **Teaching**

### **Courses Taught at ASU:**

1. EGR 280: Engineering Statistics (sophomore level). Spring 2019 - present
2. EGR 218: Materials and Manufacturing Processes (sophomore level). Summer 2022.
3. MFG 598/MFG573: Micro/Nano Additive Manufacturing (senior & graduate level). Fall 2018 - present
4. ASU 101: The ASU Experience. Fall 2022.